

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0238

Bandera Road 345 kV
December 04, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0238. The Interconnection Customer proposes a 659.1 MW natural gas combined cycle facility interconnecting at the Bandera Road 345 kV substation in the Pennsylvania area. The request was received on May 17, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 164 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,083,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Bandera Rd / Owatonna 345 KV line, the Bandera Rd / Cherokee Falls Substation 138 KV L loads to 170.8% of its 280 MVA rating. Project DFAX: 66.0%.

Bandera Rd Substation 345/138kV Auto: 121.2% loading (544 MVA) following the outage of the Bandera Rd / Owatonna 345 KV; DFAX = 100.0%.

Thetford Substation / Bowling Green Peaking 138 KV: 108.2% loading (221 MVA) following the outage of the Bandera Road / Owatonna 345 KV; DFAX = 11.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bandera Rd / Cherokee Falls Substation 138 KV L	Rebuild line with higher-capacity conductor	\$1,473,000	100% (\$1,473,000)
Bandera Rd Substation 345/138kV Auto	Replace transformer with higher-rated unit	\$13,207,000	100% (\$13,207,000)
Bowling Green Peaking / Thetford 138 KV	Reconductor line	\$403,000	100% (\$403,000)
	Total allocated to project		\$15,083,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.